

Software Engineering 2

Workshop - 2
Part - 1

Arnold Oswaldo Acosta Ortega
Omar Alejandro Salazar Carvajal
Juan David Peña Lozada
Luis Rodrigo Rivera Rivera
Nicolas David Moreno Villanueva

Systems and Computer Engineering
National University of Colombia
Bogotá DC October 2026

1. System domain **3**
1.1 System Context 3
1.1.1 System Description 3
1.1.2 System Objectives 3
1.1.3 System Boundary 4
1.1.4 External Entities and Interactions 4
1.1.5 Information Entering and Leaving the System 5
1.1.6 Assumptions and Exclusions 5
1.2. Actor Characterization 6
1.3. Business Rules 7
1.3.1 Account and Access Rules 7
1.3.2 Learning and Evaluation Rules 8
1.3.3 Budget Planning Rules These are among the most important rules because they determine the mathematical validity of the application's principal planning feature. 9
1.3.4 Financial Simulation Rules 10
1.3.5 Administration and Content Rules 10
1.3.6 AI and Data-Protection Rules 11
2 CRC Cards **12**
2.1Identification and Description of Main Classes 12
2.2 Specification of Responsibilities and Collaborators 12
2.3 CRC Cards (Table Format - One Per Class) 13
3. Class Diagram **17**
4. Relational Database Model **18**
4.1 Profile and gamification 18
4.2 Educational module and assessments 18
4.3 Financial simulation 18
4.4 Budget planning and AI 18
5. Mockups **19**

1. System domain

1.1 System Context

1.1.1 System Description

FinLearn is a web application designed to teach users personal finance through short lessons, interactive exercises, financial decision-making simulations, and monthly budgeting tools. Its purpose is to help users understand financial concepts, practice responsible financial decisions in a simulated environment, and organize the distribution of their monthly income.

The system provides educational content about savings, debt, budgeting, financial products, and compound interest. It also enables users to track their learning progress through experience points (XP), levels, activity streaks, and badges.

FinLearn includes an optional external artificial intelligence (AI) service that explains a user's budget in accessible language, subject to the user's explicit consent.

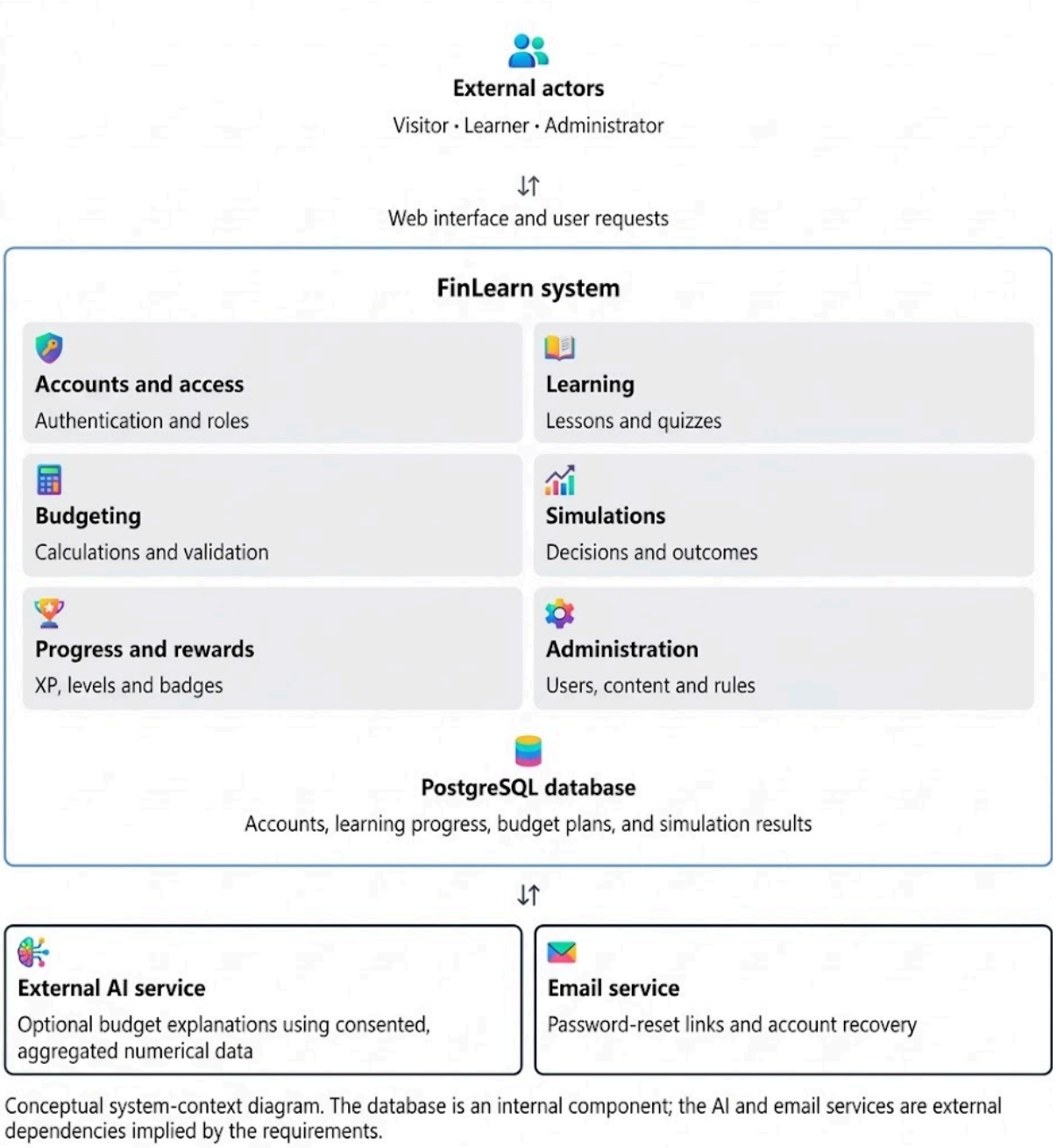
A fundamental boundary of the system is that FinLearn is an educational application, not a banking platform. It does not transfer money, connect to bank accounts, or execute real financial transactions. Budgets and financial scenarios are simulations intended for learning purposes.

1.1.2 System Objectives

The main objectives of FinLearn are:

- Financial education: Provide organized learning paths, short lessons, quizzes, and immediate feedback.
- Budget planning: Calculate a monthly budget based on income, fixed expenses, savings goals, and distribution rules.
- Decision-making practice: Simulate the consequences of spending, saving, and debt-related decisions without real financial risk.
- Progress tracking: Save learning results, budget plans, simulation histories, XP, levels, streaks, and badges.
- Platform administration: Allow authorized administrators to manage users, educational content, distribution rules, and simulation scenarios.
- Personalized explanations: Use an external AI service, when authorized, to explain budget distributions while protecting personal information.

1.1.3 System Boundary



Conceptual system-context diagram. The database is an internal component; the AI and email services are external dependencies implied by the requirements.

1.1.4 External Entities and Interactions

Entity or system	Interaction with FinLearn
Visitor	Creates an account using a display name, email, and password.

Entity or system	Interaction with FinLearn
Learner	Studies lessons, answers quizzes, manages budgets, completes simulations, and checks progress.
Administrator	Manages user accounts, content, distribution rules, and simulation configurations.
External AI service	Receives permitted aggregated numerical data and returns an explanation of the budget.
Email service	Delivers temporary password-reset links.
Web browser	Provides the interface through which users access the application.

1.1.5 Information Entering and Leaving the System

Inputs

- Account details and authentication credentials.
- Learning selections and quiz answers.
- Monthly net income in Colombian pesos (COP), fixed expenses, expense categories, and optional savings goals.
- Budget distribution rules selected by the learner.
- Choices made in simulated financial scenarios.
- Administrative updates to learning materials and simulation parameters.
- Explicit consent for optional AI explanations.

Outputs

- Authentication results and account-recovery confirmations.
- Lessons, quizzes, scores, and educational feedback.
- Calculated budget allocations, remaining balances, charts, and deficit alerts.
- Simulation indicators, consequences, final reports, and educational outcomes.
- Learning progress, XP, levels, streaks, and badges.
- Optional AI-generated budget explanations.

1.1.6 Assumptions and Exclusions

- The application is web-based.
- PostgreSQL is the selected database.

- Financial values are entered for educational budgeting and simulation purposes.
- The system does not access bank credentials or card details.
- The external AI integration is optional and requires explicit authorization.
- Performance, availability, recovery, and usability targets are preliminary assumptions for an academic pilot and require validation.

1.2. Actor Characterization

1.2.1 Visitor

Unauthenticated user

Description: A person who accesses FinLearn without having a registered account.

Goals: Create an account and gain access to the educational features of the platform.

Main interactions:

- Enter a display name, email address, and password.
- Submit a registration request.
- Receive confirmation or an error if the email is already registered.

The visitor can use the registration and account-recovery entry points but does not have access to protected learner features until authenticated.

1.2.2 Learner (**Apprentice**)

Primary actor

Description: A registered user older than 13 years old spanish speaker, who wants to improve their financial knowledge and practice personal financial planning.

Goals: Learn financial concepts, create a viable budget, practice financial decisions, and monitor educational progress.

Main interactions:

- Log in, update their profile, and change their password.
- Study learning paths and micro-lessons.
- Answer quizzes and review feedback.
- Create, calculate, save, and review budget plans.
- Participate in simulations and inspect their consequences.
- View XP, levels, streaks, badges, and historical results.
- Request an AI-generated explanation of a budget after providing consent.

The learner manages their own profile and financial plans. They cannot perform administrative operations or execute actual banking transactions through FinLearn.

1.2.3 Administrator

Privileged user

Description: An authorized person responsible for maintaining and operating the platform's content, accounts, and educational configurations.

Goals: Keep learning materials accurate, manage access, and configure budget rules and simulation scenarios.

Main interactions:

- Consult user accounts and modify permitted roles.
- Deactivate and reactivate accounts.
- Create, edit, publish, and unpublish educational content.
- Manage lessons, questions, and answer options.
- Create and maintain budget distribution rules.
- Configure simulation scenarios and their expected consequences.

Administrative permissions must be checked on the application server. The administrator must never be able to view or recover users' original passwords.

1.3. Business Rules

The rules below are derived from the project's functional requirements and acceptance criteria. Where the project does not specify an exact threshold or policy, the rule is identified as a proposal that requires confirmation.

1.3.1 Account and Access Rules

ID	Business rule	Basis
BR-01	Every registered account must have a unique email address. Duplicate email registrations must be rejected.	RF-01
BR-02	Only authenticated users with an active account may access protected learner features.	RF-02, RF-05
BR-03	The system must enforce permissions according to the user's assigned role: learner or administrator.	RF-02.2

ID	Business rule	Basis
BR-04	Password recovery must use a temporary link that expires and cannot be reused.	RF-03 and HU 1.3
BR-05	Passwords must be stored as secure hashes. Administrators must not be able to retrieve original passwords.	RF-05, RNF-04
BR-06	Deactivating an account must prevent access while preserving its existing data and progress.	RF-05 and HU 1.5
BR-07	Changing a learner's profile information must not detach their saved budgets or educational progress from their account.	RF-04

1.3.2 Learning and Evaluation Rules

ID	Business rule	Basis
BR-08	Learning paths must be organized by levels and modules covering the configured financial topics.	RF-06
BR-09	A learner may access a restricted lesson only after satisfying its configured prerequisites.	HU 3.1–3.2
BR-10	Quiz answers must be evaluated against the configured correct answers, and feedback must explain incorrect responses.	RF-08.1
BR-11	A learner's completed lessons, quiz results, and achieved level must be saved and associated with their account.	RF-07.1, RF-09
BR-12	XP and badges must be awarded only when their predefined conditions are satisfied. The same unique event must not award the same reward twice.	RF-18
BR-13	Activity streaks must be calculated according to predefined daily continuity rules.	RF-18

We did not specify the exact passing score for every quiz, the number of XP awarded per activity, or the precise thresholds for levels and badges. These values should be configurable or formally agreed upon before implementation.

1.3.3 Budget Planning Rules These are among the most important rules because they determine the mathematical validity of the application's principal planning feature.

ID	Business rule	Basis
BR-14	Monthly net income and expense values must not be negative.	RF-11
BR-15	Each budget plan must use an available distribution rule published by the administrator.	RF-12
BR-16	The sum of all percentages in a distribution rule must equal exactly 100% before the rule can be applied.	RF-12.1
BR-17	Each category allocation must be calculated from monthly net income and the percentage assigned to that category.	RF-13
BR-18	Fixed expenses must be deducted from the allocation of their corresponding categories to calculate the remaining balance.	RF-13.1
BR-19	If total fixed expenses exceed monthly net income, the plan must be flagged as having a deficit and cannot be saved as balanced.	RF-14, RF-14.1
BR-20	If fixed expenses exceed the allocation of a particular category, the system must identify that category as financially unviable.	RF-14
BR-21	A plan must display its allocated amounts and percentages numerically and graphically.	RF-15
BR-22	Learners must be able to save budget-plan versions and consult their previous saved plans.	RF-15.1
BR-23	Savings goals are optional; if supplied, they must be incorporated into the budget calculations according to the selected rule and the agreed planning logic.	Scope, page 3

1.3.4 Financial Simulation Rules

ID	Business rule	Basis
BR-24	Each simulation decision must be selected from the options configured for the current scenario.	HU 4.1
BR-25	Every valid decision must apply its configured consequences to the relevant simulation variables.	RF-16.1
BR-26	The simulator must update the indicators affected by a decision and leave unaffected indicators unchanged.	HU 4.3
BR-27	When an indicator reaches its configured critical threshold, the system must display a visual warning.	HU 4.4
BR-28	A completed 30-day simulation must generate a final report with the outcome and any applicable rewards.	HU 4.5
BR-29	When a configured financial-disaster condition occurs, the simulation must stop and display the disaster screen.	HU 4.6
BR-30	Restarting a simulation after a financial disaster must start a new cycle from day one.	HU 4.6
BR-31	Administrators must be able to modify scenario parameters and expected consequences without changing the application's source code.	RF-17

1.3.5 Administration and Content Rules

ID	Business rule	Basis
BR-32	Only authorized administrators may create, modify, publish, or unpublish educational content.	RF-10
BR-33	Only published content may be offered as available content to learners, subject to access prerequisites.	HU 5.3

ID	Business rule	Basis
BR-34	Administrators must manage distribution rules and simulation scenarios through the administration interface.	RF-17
BR-35	Changes to published rules and scenarios must be reflected in subsequent budget calculations or new simulations as applicable.	RNF-11, HU 5.4
BR-36	Administrative actions must be authorized by server-side role checks, not merely by hiding interface options.	RNF-04

1.3.6 AI and Data-Protection Rules

ID	Business rule	Basis
BR-37	The system may request an AI-generated budget explanation only after the learner explicitly authorizes the operation.	RF-20
BR-38	The AI service must receive only the aggregated numerical data needed for the explanation, without personally identifiable information.	RF-20, RNF-05
BR-39	AI-generated explanations must be clearly labeled as AI-generated.	RF-20
BR-40	If the AI service fails or takes longer than the specified 10-second limit, FinLearn must preserve the budget calculation and offer a predefined explanation.	RNF-10, RF-20
BR-41	Learners' personal data must be accessible only to the account holder and authorized personnel according to their roles.	RNF-05
BR-42	FinLearn must not request bank credentials or card details, and must not execute real financial transactions.	Scope, page 3

2 CRC Cards

2.1Identification and Description of Main Classes

In accordance with the domain model and functional scope of FinLearn, the following six primary classes have been identified to fulfill system operations, user interactions, and business rules:

1. User: Represents any registered entity interacting with the FinLearn platform. Holds basic profile details (display name, email), account status (active/deactivated), and role privileges (Visitor, Learner, Administrator).

2. AuthenticationSystem: Responsible for security, credential verification, session lifecycle management (login/logout), password hashing enforcement (BR-05), and role-based access control (BR-06).

3. BudgetPlan: Encapsulates educational financial budgeting. Manages income, fixed and variable expenses, balance calculation, percentage distributions (e.g., 50/30/20 rule), non-negative validation (BR-14), and plan persistence.

4. SimulationScenario: Handles interactive financial decision scenarios. Tracks situational indicators, applies user choices, prevents tampering with previous decision steps (BR-26), and evaluates overall performance scores.

5. LearningRoute: Organizes financial learning content into structured sequential routes and micro-lessons. Manages lesson progress tracking and navigation for learners (HU 3.1).

6. AIExplainer: Interacts with external or integrated AI services to generate tailored educational explanations and budget feedback, ensuring feedback is only requested after full budget persistence (BR-37).

2.2 Specification of Responsibilities and Collaborators

Each class in the FinLearn system carries specific responsibilities derived from system business rules (BR-05 to BR-37) and functional requirements (RF-01 to RF-15). The table below outlines these responsibilities along with the collaborating classes required to fulfill them:

Class Name	Key Responsibilities	Collaborators
User	• Store profile data (name, email) • Maintain state (active/inactive) • Role authorization context	AuthenticationSystem, BudgetPlan, SimulationScenario, LearningRoute

	• Track budget & route history	
AuthenticationSystem	• Validate login credentials • Hash passwords securely (BR-05) • Enforce admin restrictions (BR-06) • Manage active user sessions	User
BudgetPlan	• Validate non-negative numbers (BR-14) • Calculate totals & net balances • Apply 50/30/20 distribution rule • Persist budget data	User, AIExplainer
SimulationScenario	• Present financial choice scenarios • Update dynamic indicators • Lock completed decision steps (BR-26) • Compute final score	User
LearningRoute	• Manage sequential micro-lessons • Record learner completion status • Serve educational route content	User
AIExplainer	• Request AI summary of budget • Verify budget persistence (BR-37) • Generate financial insights	BudgetPlan

2.3 CRC Cards (Table Format - One Per Class)

Below are the detailed CRC Cards presented in standard tabular format, providing one individual CRC card per identified class:

Class: User	
Responsibilities	**Collaborators**
• Maintain user account information (display name, email address, password hash). • Store account status flag (active vs. deactivated). • Hold user role classification (Visitor, Learner, Administrator). • Maintain references to created budget plans, active simulation progress, and completed micro-lessons.	• AuthenticationSystem • BudgetPlan • SimulationScenario • LearningRoute

Class: AuthenticationSystem	
Responsibilities	**Collaborators**
• Authenticate credentials entered during user login. • Manage user sessions (login initiation, token management, session expiration, and logout). • Ensure secure password hashing before persistence (Business Rule BR-05). • Enforce administrative privileges and account reactivation rules (Business Rule BR-06).	• User

Class: BudgetPlan	
Responsibilities	**Collaborators**
• Store monthly net income, fixed expenses, and variable expenses. • Enforce business validation: income and expense values must not be negative (Business Rule BR-14). • Calculate total expenses, net balance, and expense percentage distributions. • Apply financial heuristics such as the 50/30/20 budgeting rule (Business Rule BR-15). • Persist created budgets for long-term user tracking and AI analysis.	• User • AIExplainer

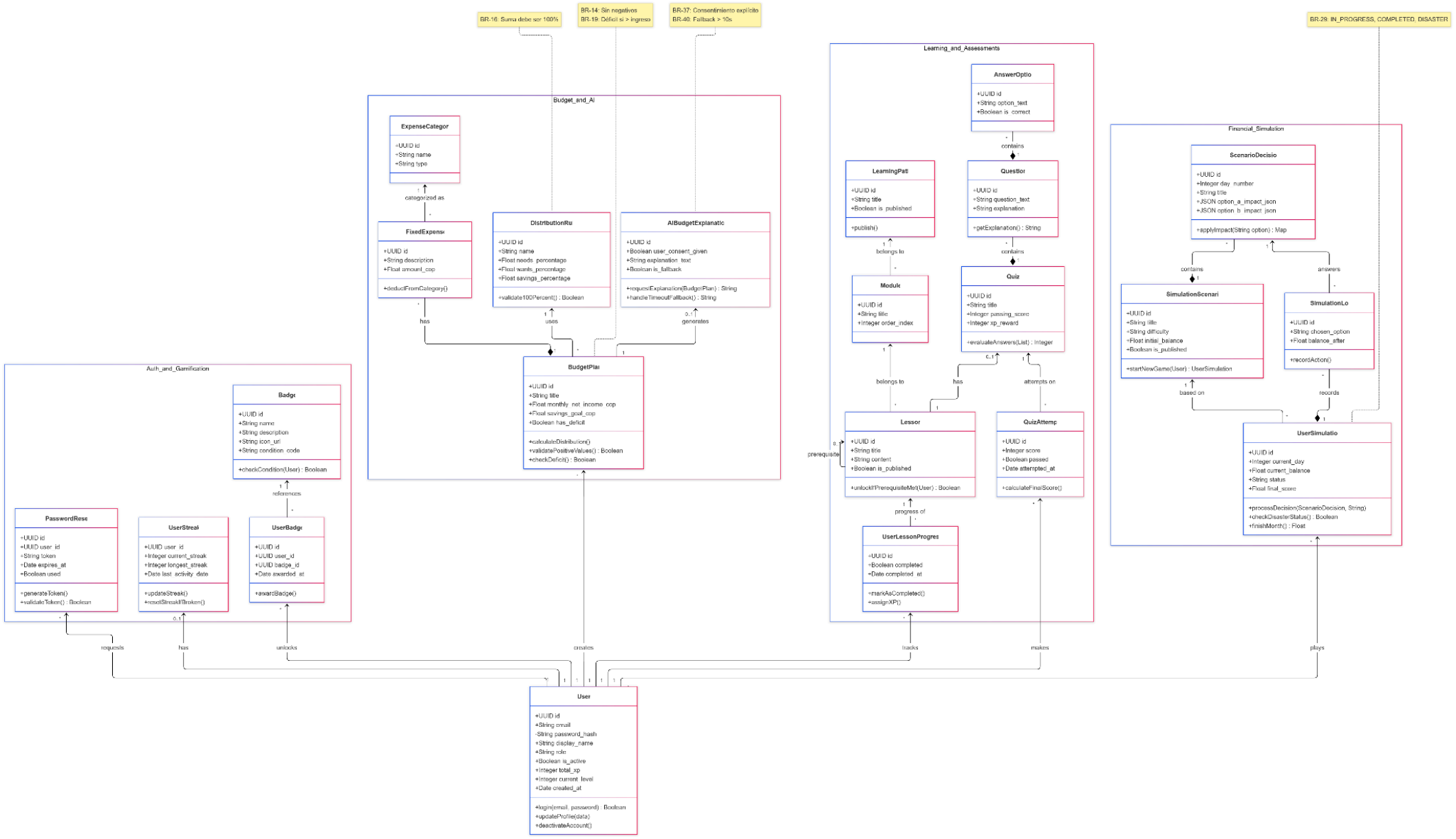
Class: SimulationScenario	
Responsibilities	**Collaborators**
• Present financial decision scenarios and decision options to the user. • Dynamically update economic indicators based on choices made by the user. • Lock previous decision steps and prevent modifying past indicators after confirmation (Business Rule BR-26). • Compute end-of-scenario summary metrics and user score.	• User

Class: LearningRoute	
Responsibilities	**Collaborators**

• Structure and manage sequential routes of educational micro-lessons. • Track user completion progress across lessons and modules. • Provide learning content and navigational flow for financial education.	• User

Class: AIExplainer	
Responsibilities	**Collaborators**
• Receive complete budget data payloads to formulate prompt contexts. • Verify that the target budget plan has been fully saved prior to issuing AI requests (Business Rule BR-37). • Generate AI-driven financial feedback, advice, and structural explanations for the user.	• BudgetPlan

3. Class Diagram



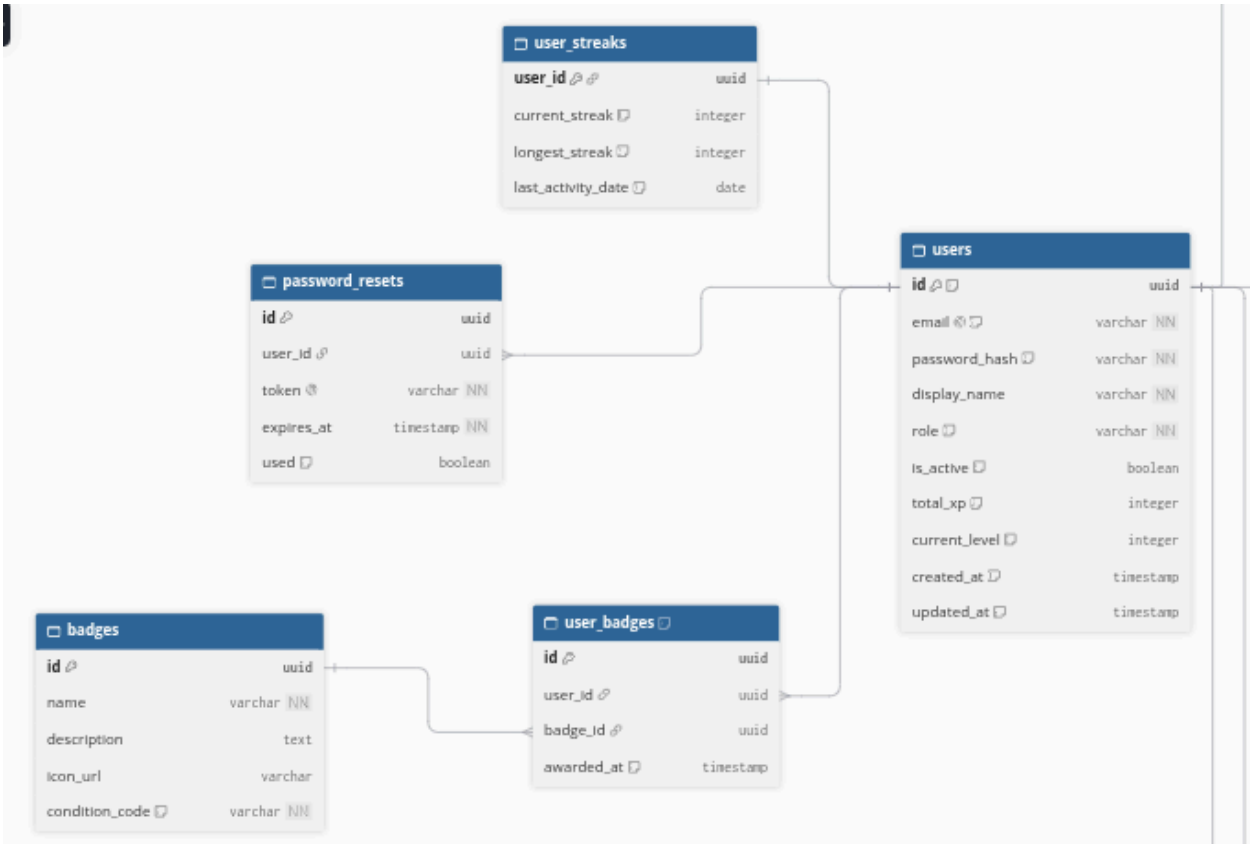
you can visualize complete in the next link:
https://mermaid.ai/app/projects/c83bbdb2-ecbc-4bed-95a6-60a977036adc/diagrams/07325da2-a84b-49bb-acbe-ecb656bdfe83/share/invite/eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJkb2N1bWVudElEIjoiMDczMjVkYTItYTg0Yi00OWJiLWFjYmUtZWNiNjU2YmRmZTgzIiwiYWNjZXNzIjoiVmlldyIsInB1cnBvc2UiOiJzaGFyZS1pbnZpdGUiLCJpYXQiOjE3OTE5MzMwOTAsImV4cCI6MTc5NDUyNTA5MH0.sQvpmU4PYqdBBFNVs8kEpx7fRjOCEdbzJY5FQZu6AHw?entryPoint=share-modal

4. Relational Database Model

You can visualize the complete diagram in the next link:
https://dbdiagram.io/d/FINLEARN-6ad1212e064322c86963b9a2

4.1 Profile and gamification

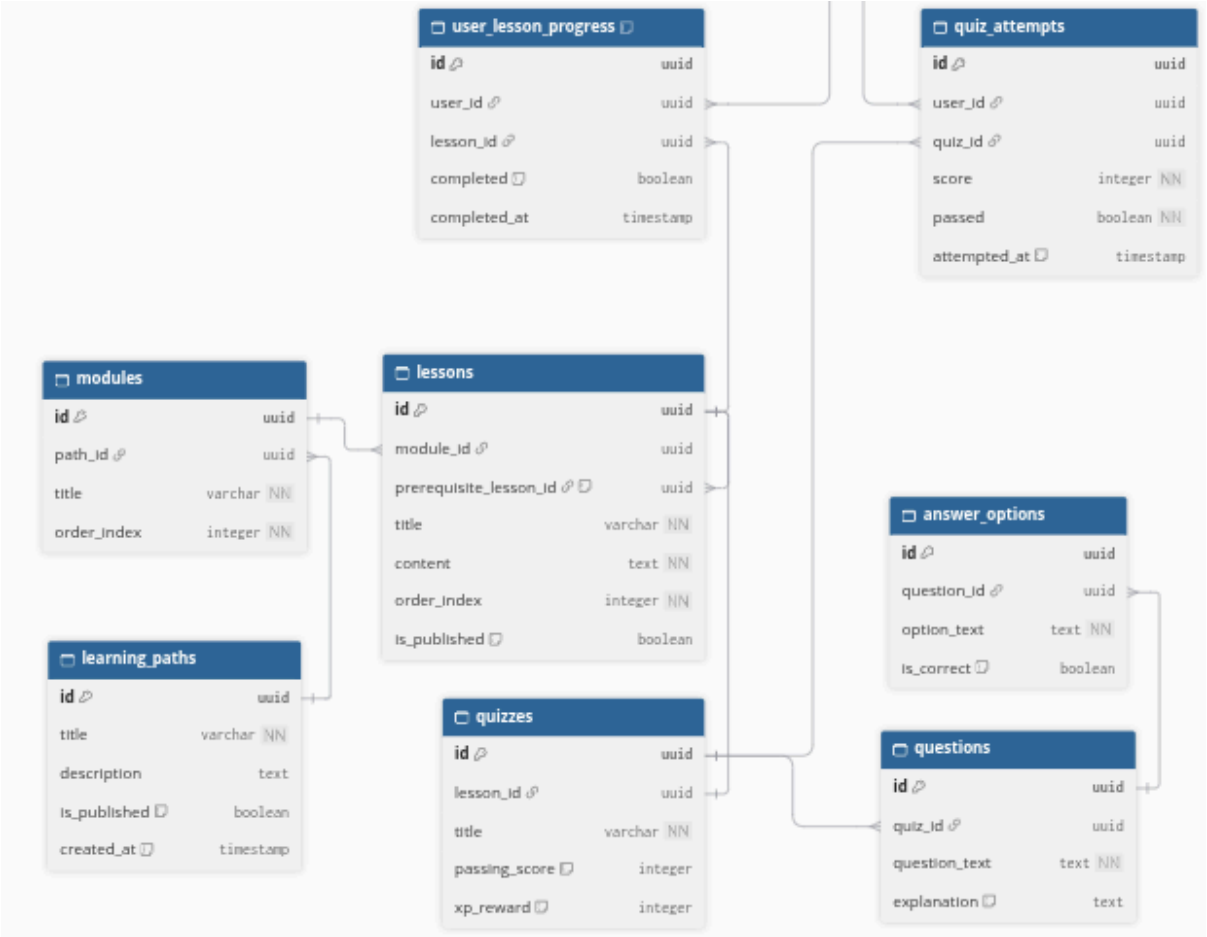
Purpose: Manages system security, access control, user authentication, and learner engagement.
Key Components: Stores user account profiles (users), enforces role-based permissions (LEARNER and ADMIN), handles secure hashed passwords, and manages temporary account recovery links (password_resets). Additionally, it drives user retention through gamification mechanics: earning unlocked badges (badges, user_badges), accumulating experience points (XP) and levels, and tracking daily activity streaks (user_streaks).



4.2 Educational module and assessments

Purpose: Structures learning content and evaluates student knowledge and progress.

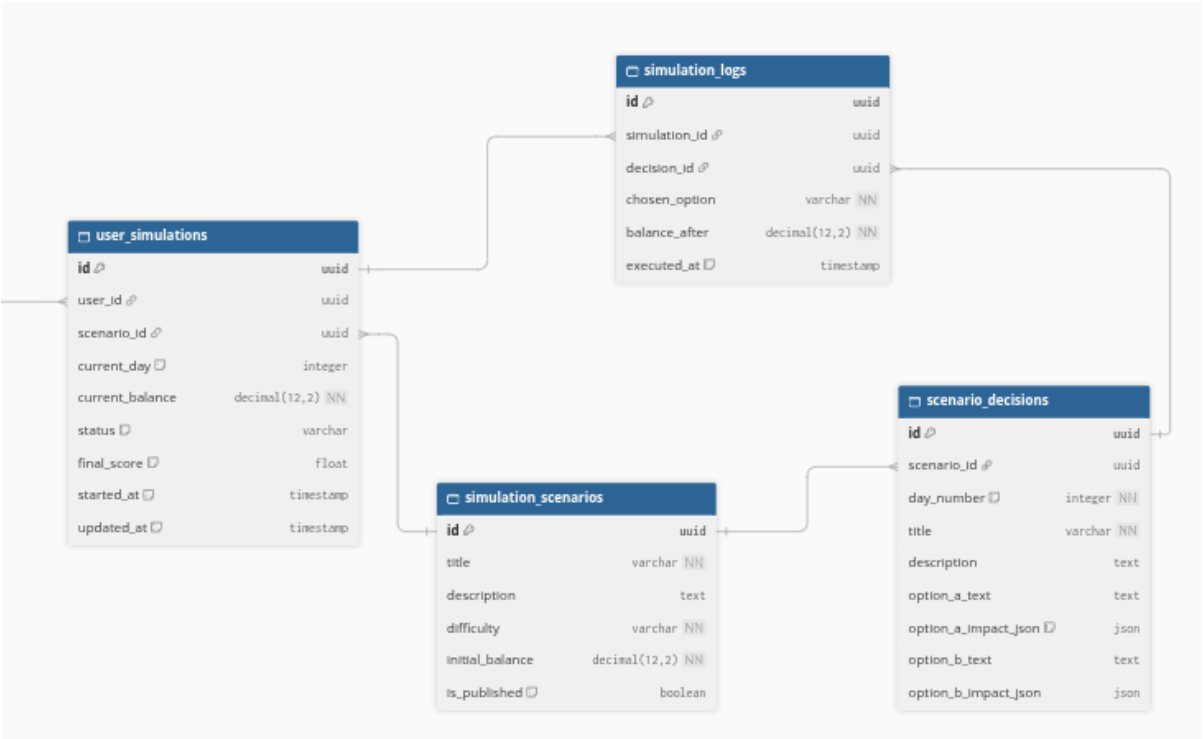
Key Components: Organizes educational material in a hierarchical tree (learning_paths → modules → lessons). Defines prerequisites between lessons and stores quizzes (quizzes, questions, answer_options) alongside answer feedback. Finally, it records each user's completed lessons (user_lesson_progress) and their quiz scores across multiple attempts (quiz_attempts).



4.3 Financial simulation

Purpose: Provides a practical, risk-free environment where learners make dynamic decisions across a 30-day simulated month without risking real money.

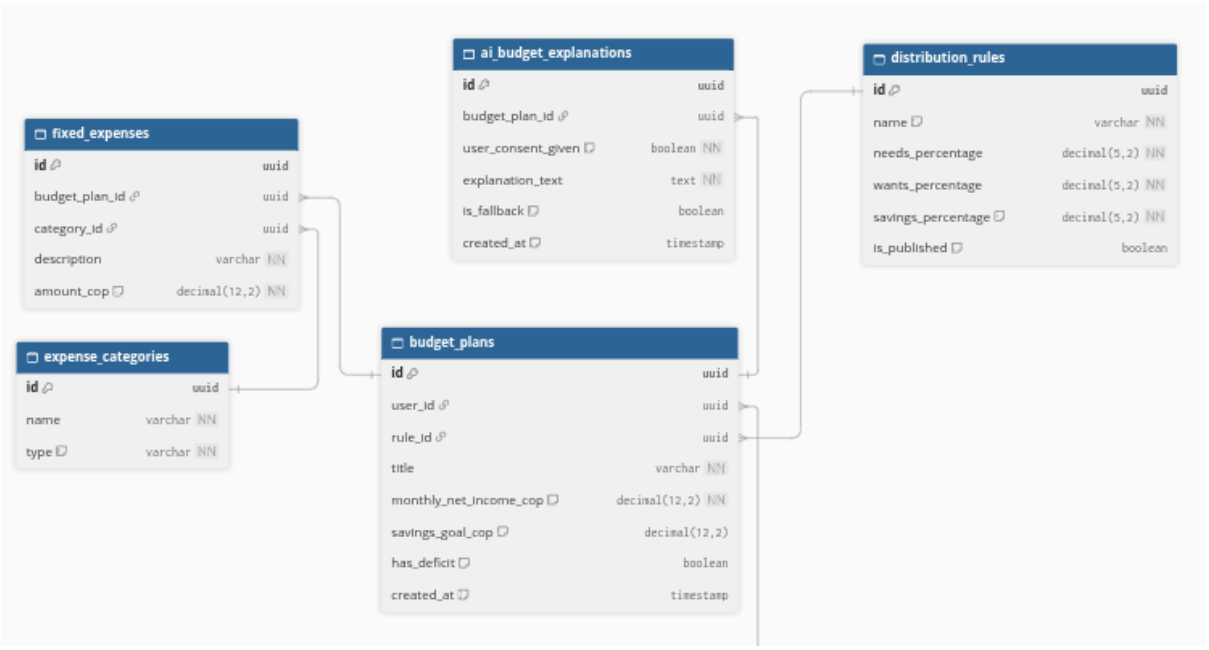
Key Components: Holds scenario templates created by administrators (simulation_scenarios) and a daily catalog of events/choices with their financial impacts (scenario_decisions). Tracks active student game sessions with current balances and financial health statuses (user_simulations), and logs every decision step-by-step (simulation_logs) to build final outcome reports or explain bankruptcy causes.



4.4 Budget planning and AI

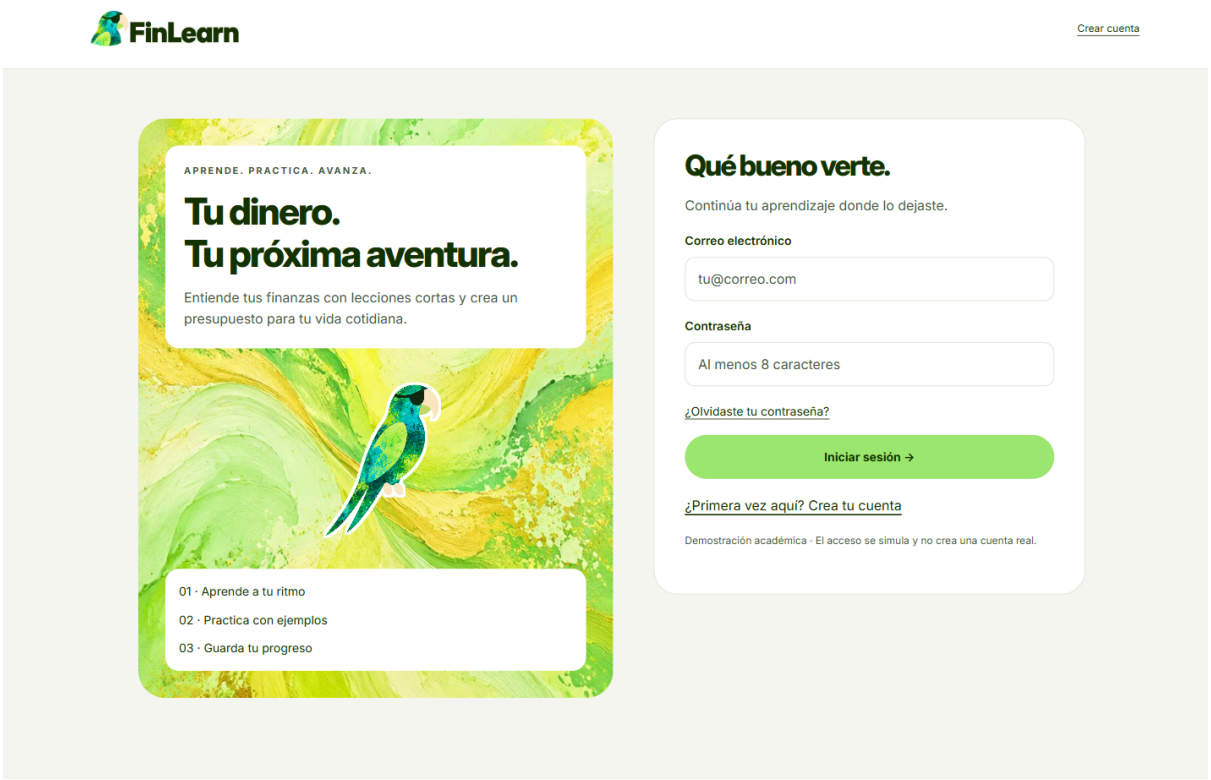
Purpose: Provides tools to structure, calculate, and diagnose monthly budgets using automated rules and optional AI insights.

Key Components: Stores budgeting allocation methodologies like the 50/30/20 rule (distribution_rules) and user budget headers (budget_plans). Connects itemized monthly costs (fixed_expenses) to spending buckets (expense_categories) to compute remaining balances and trigger deficit alerts. Lastly, it stores AI-generated financial evaluations (ai_budget_explanations), auditing explicit user consent and handling service timeout fallbacks.

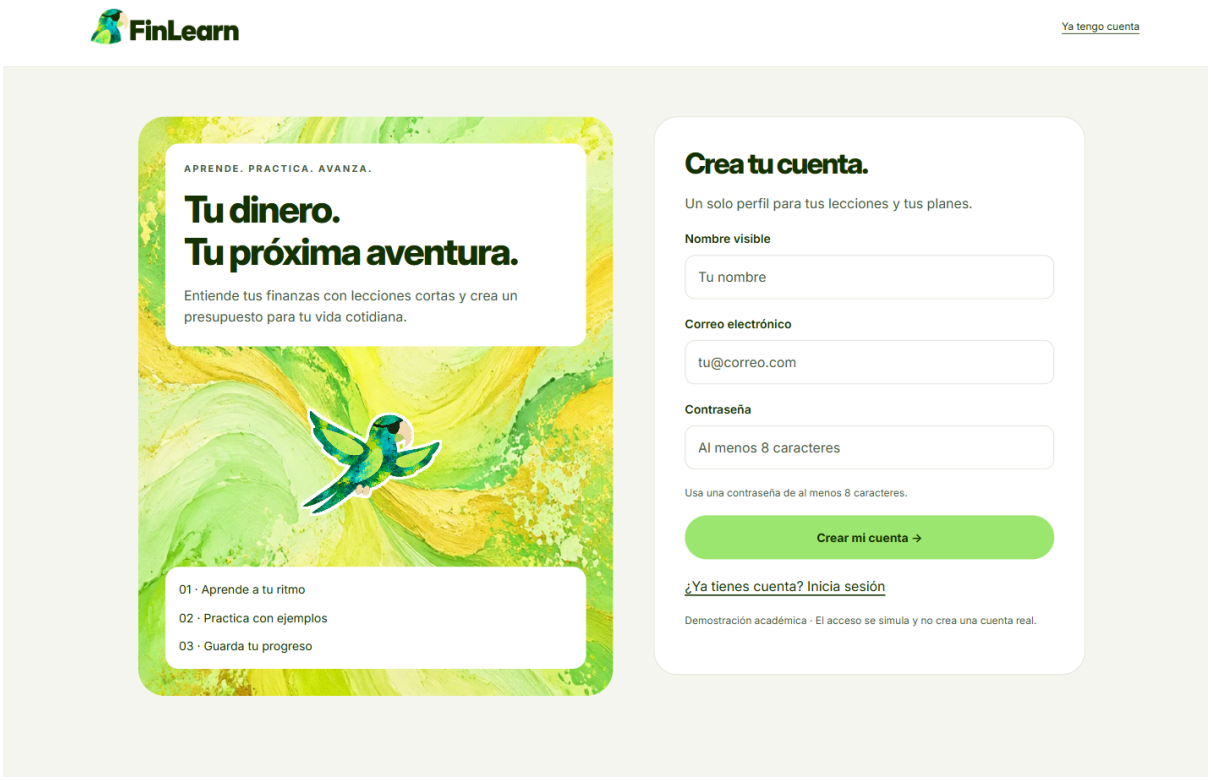


5. Mockups

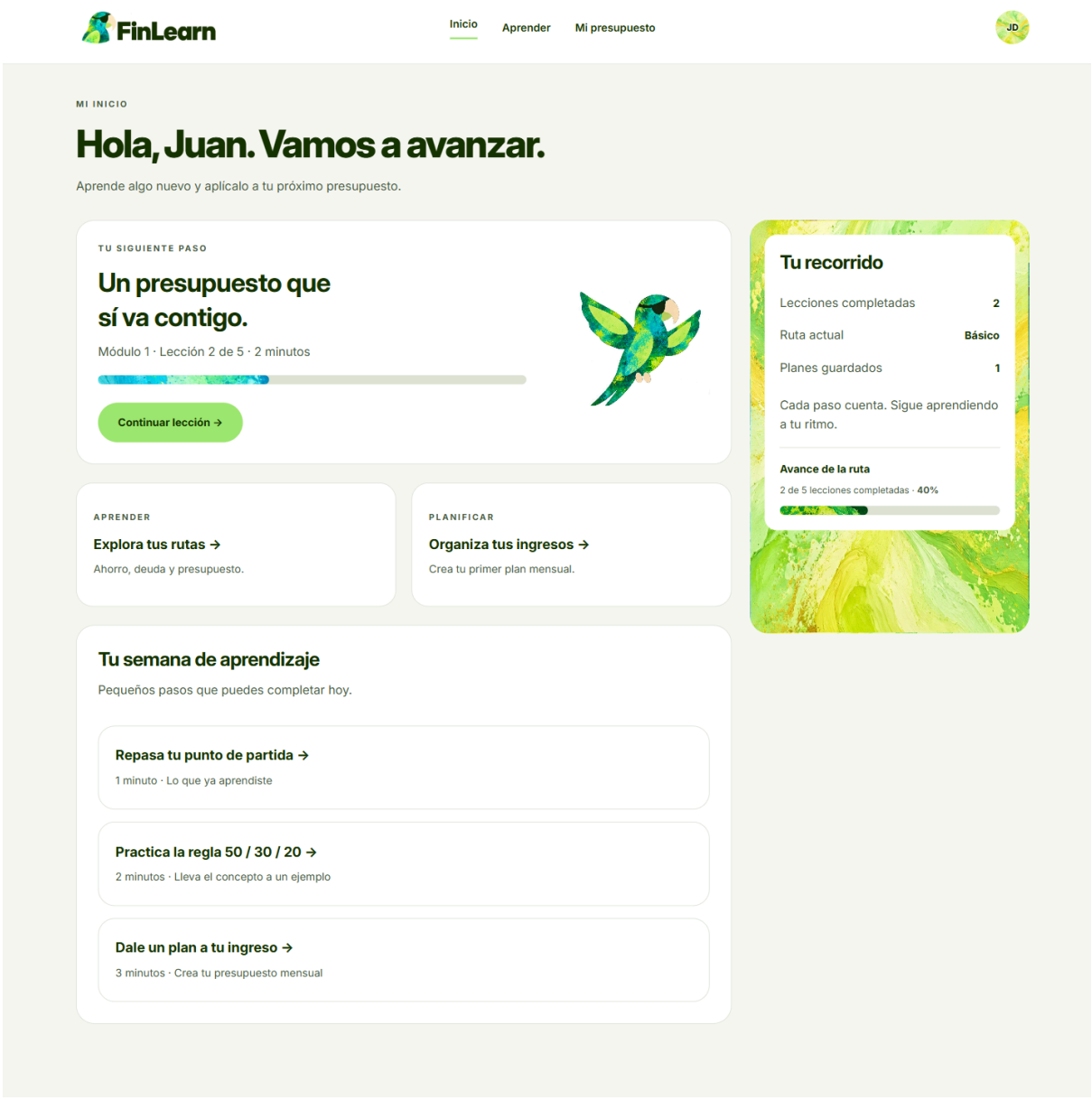
Login



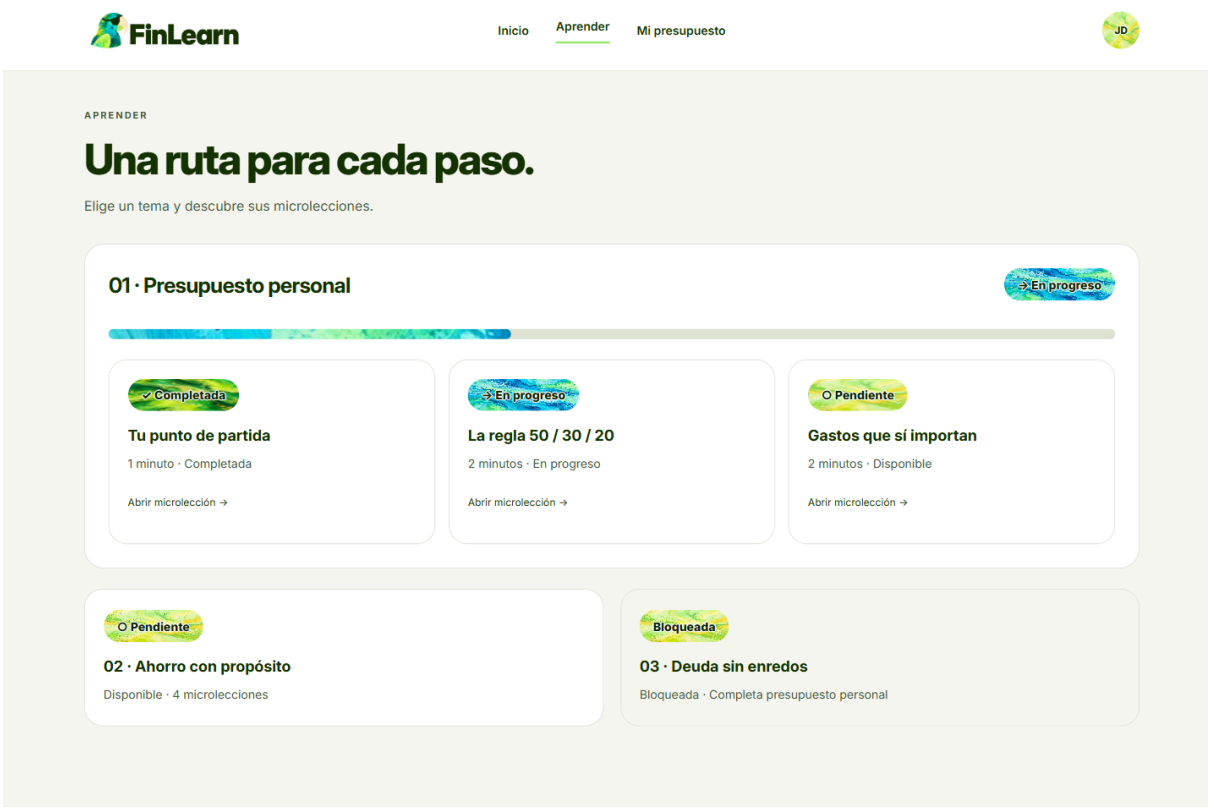
Register



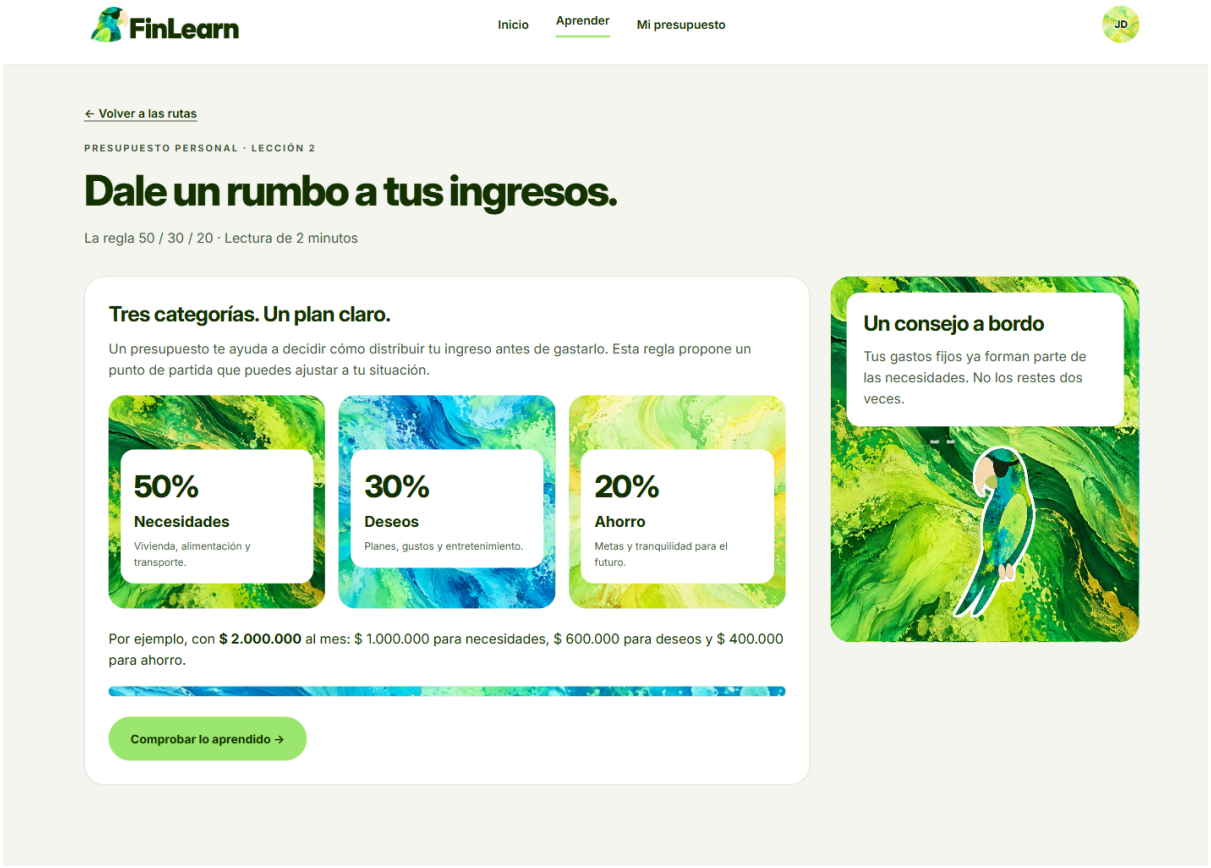
Aprentice Homescreen



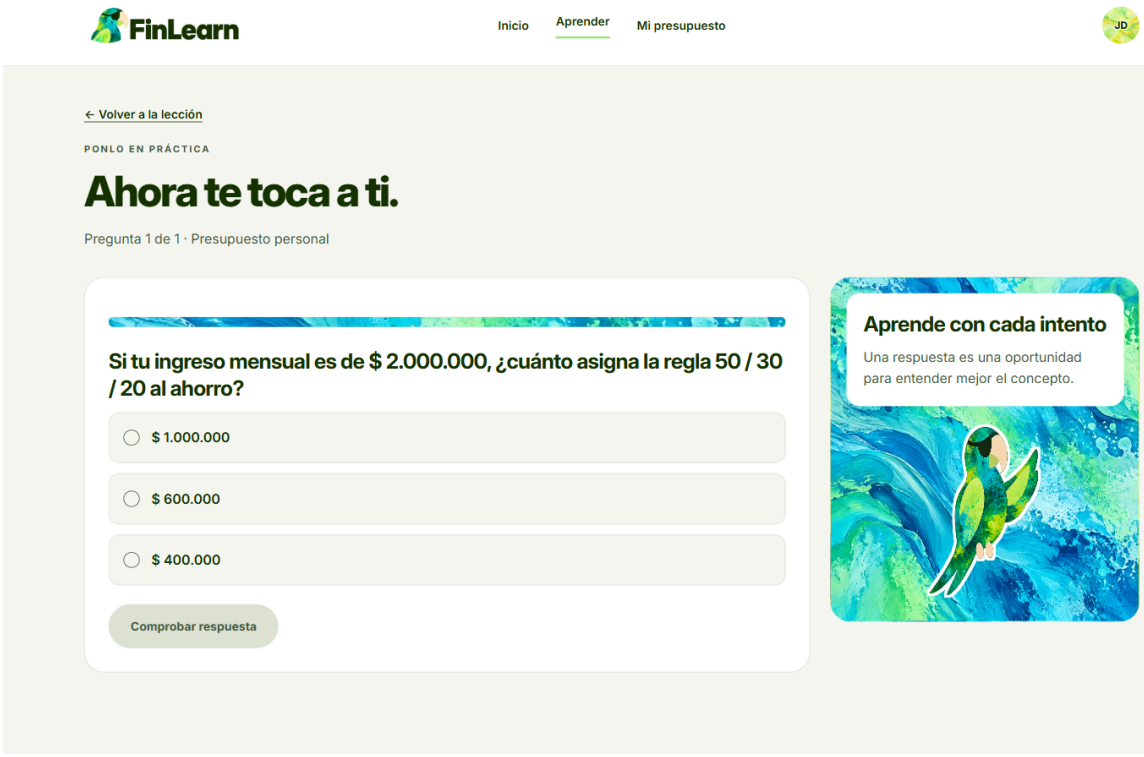
Knowledge Routes



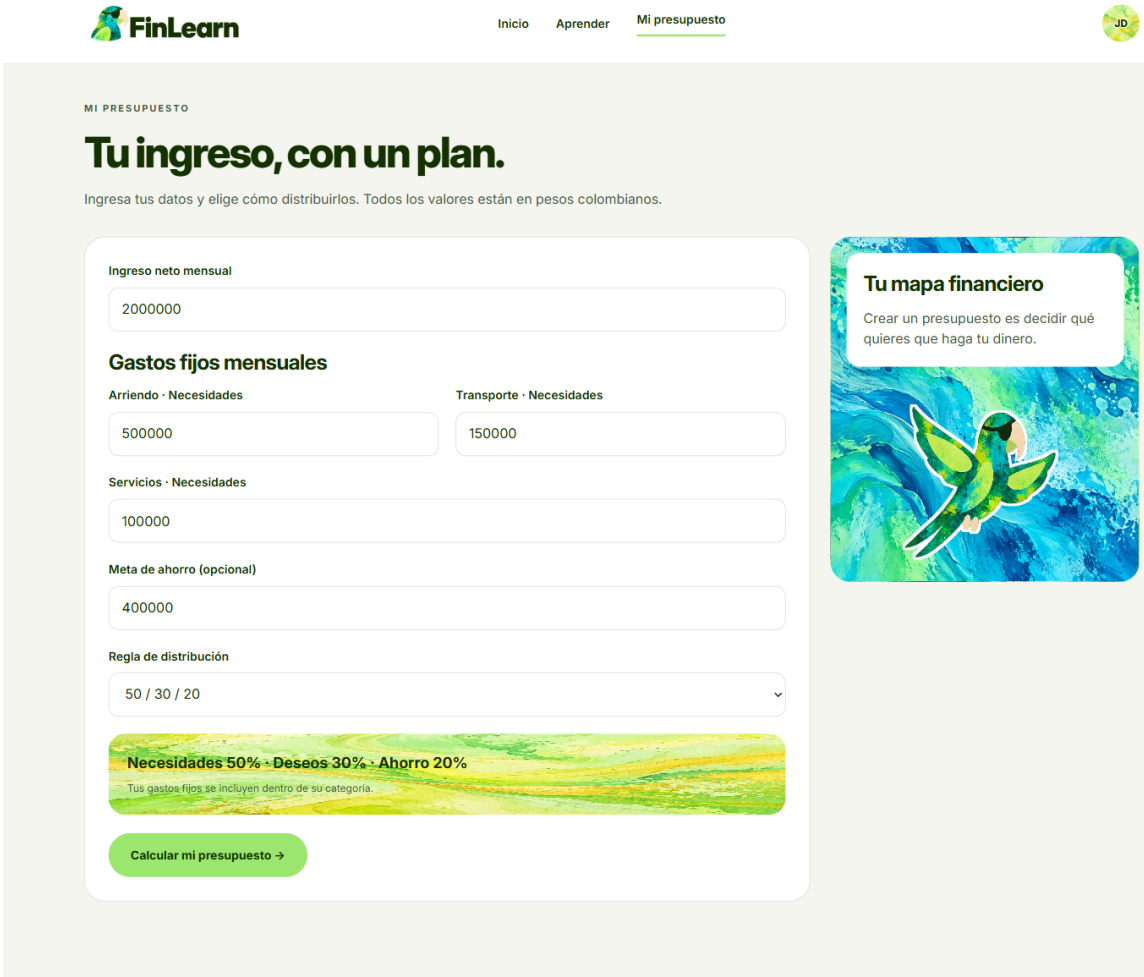
Micro-election



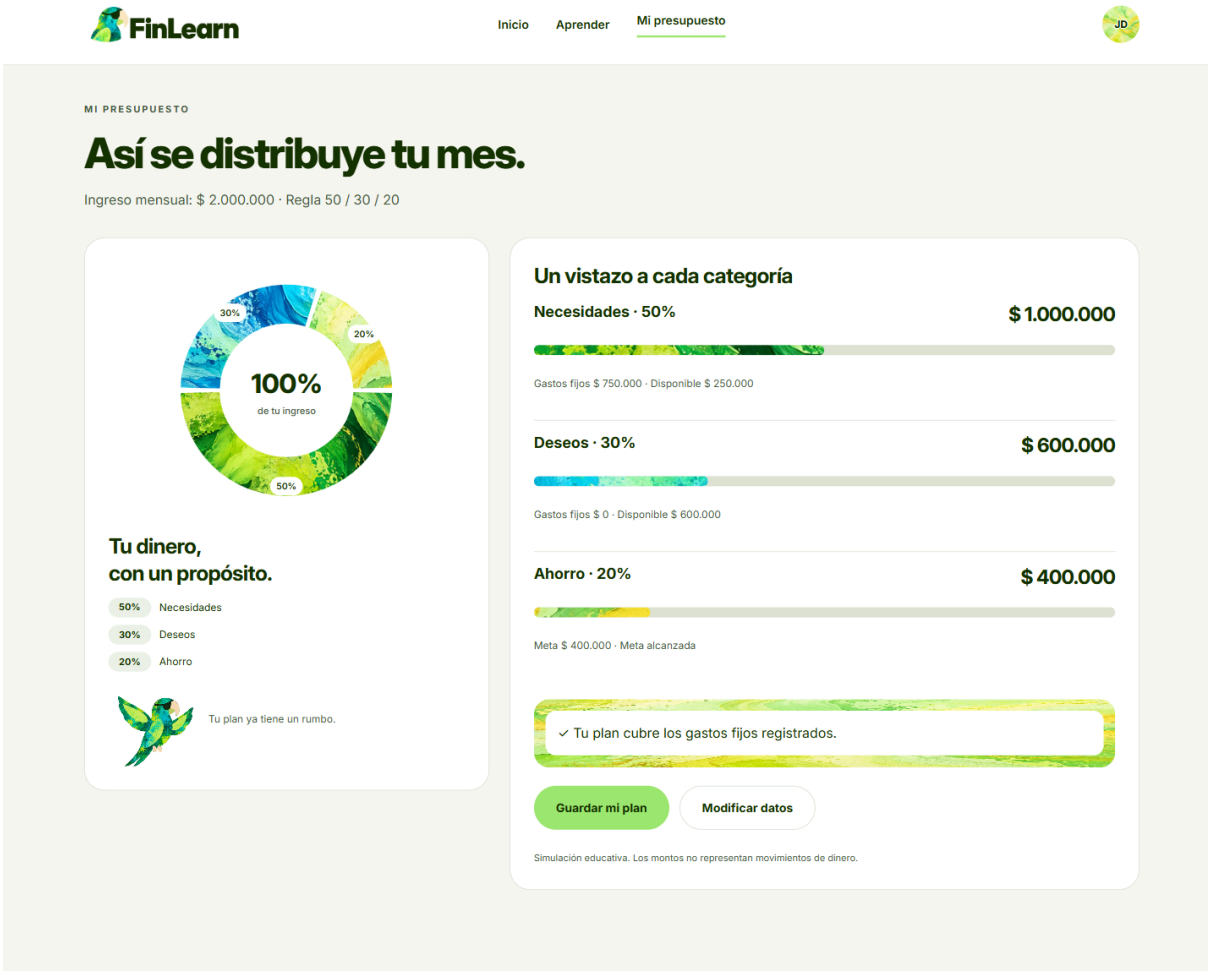
Poll



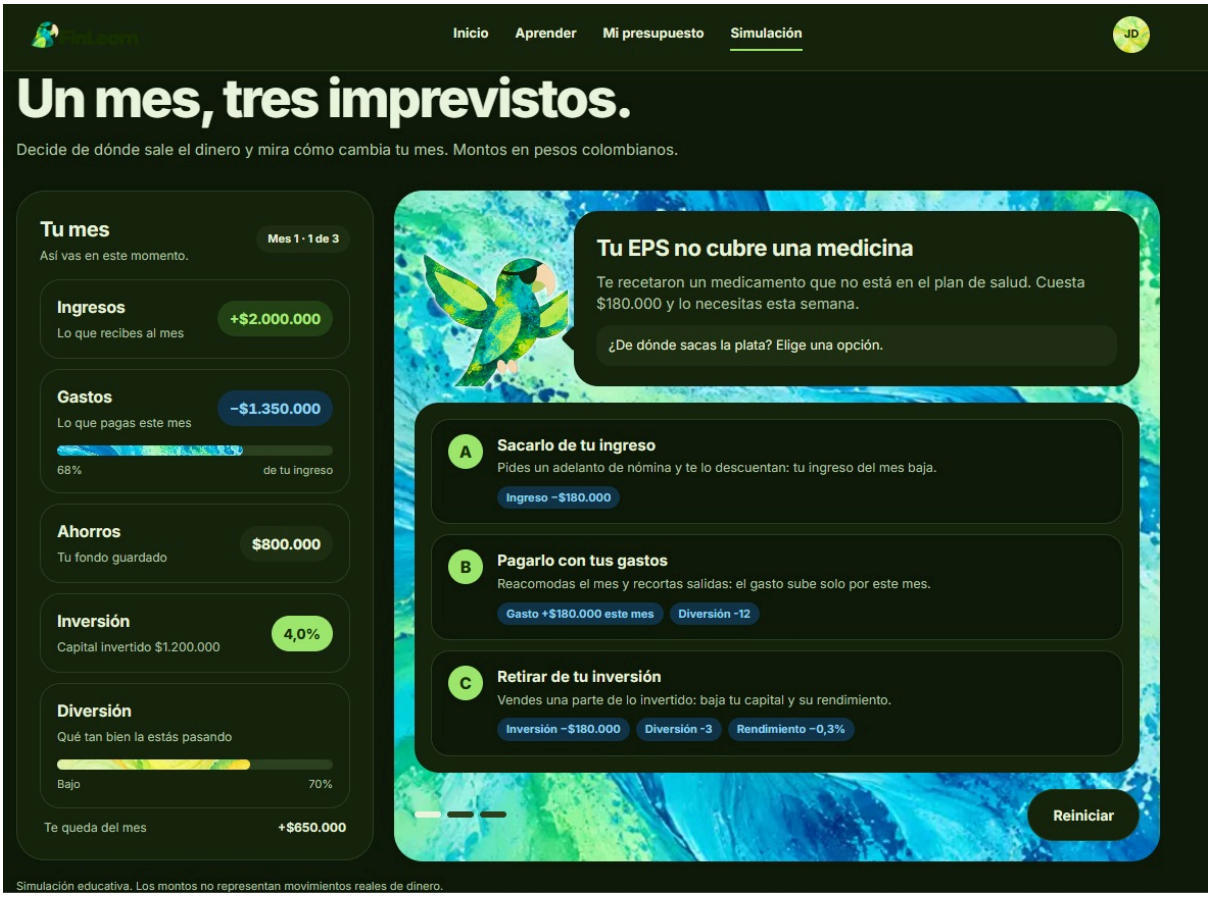
Create Budget



Budget Results



Simulación


Inicio
Aprender
Mi presupuesto
Simulación
JD
Un mes, tres imprevistos.
Decide de dónde sale el dinero y mira cómo cambia tu mes. Montos en pesos colombianos.
Tu mes
Así vas en este momento.
Mes 1 · 1 de 3
Ingresos
Lo que recibes al mes
+$2.000.000
Gastos
Lo que pagas este mes
−$1.350.000
68%
de tu ingreso
Ahorros
Tu fondo guardado
$800.000
Inversión
Capital invertido $1.200.000
4,0%
Diversión
Qué tan bien la estás pasando
Bajo
70%
Te queda del mes
+$650.000
Tu EPS no cubre una medicina
Te recetaron un medicamento que no está en el plan de salud. Cuesta $180.000 y lo necesitas esta semana.
¿De dónde sacas la plata? Elige una opción.
A
Sacarlo de tu ingreso
Pides un adelanto de nómina y te lo descuentan: tu ingreso del mes baja.
Ingreso −$180.000
B
Pagarlo con tus gastos
Reacomodas el mes y recortas salidas: el gasto sube solo por este mes.
Gasto +$180.000 este mes
Diversión -12
C
Retirar de tu inversión
Vendes una parte de lo invertido: baja tu capital y su rendimiento.
Inversión −$180.000
Diversión -3
Rendimiento −0,3%
Reiniciar
Simulación educativa. Los montos no representan movimientos reales de dinero.